TECH ZEPHYR4.0 ML HACKATHON

PROBLEM STATEMENT: 2D Sketch to a 3D Asset

TEAM NAME: BYTEFORCE **PROJECT NAME: SKETCH2SHAPE**

#OBJECTIVE:

1. 2D Sketch Input: To provide a user-friendly interface that allows users to draw sketches directly or upload existing sketch images.

2. Sketch Recognition: To use computer vision techniques to identify and understand the major shapes, contours, and structural features of the input sketch.

3. AI-Based Image Generation: To utilize Generative AI techniques to enhance and refine the sketch while preserving its original structure and visual characteristics.

4. 3D Model Reconstruction: To transform the processed 2D representation into a three-dimensional model with meaningful depth, volume, geometry, and proportions.

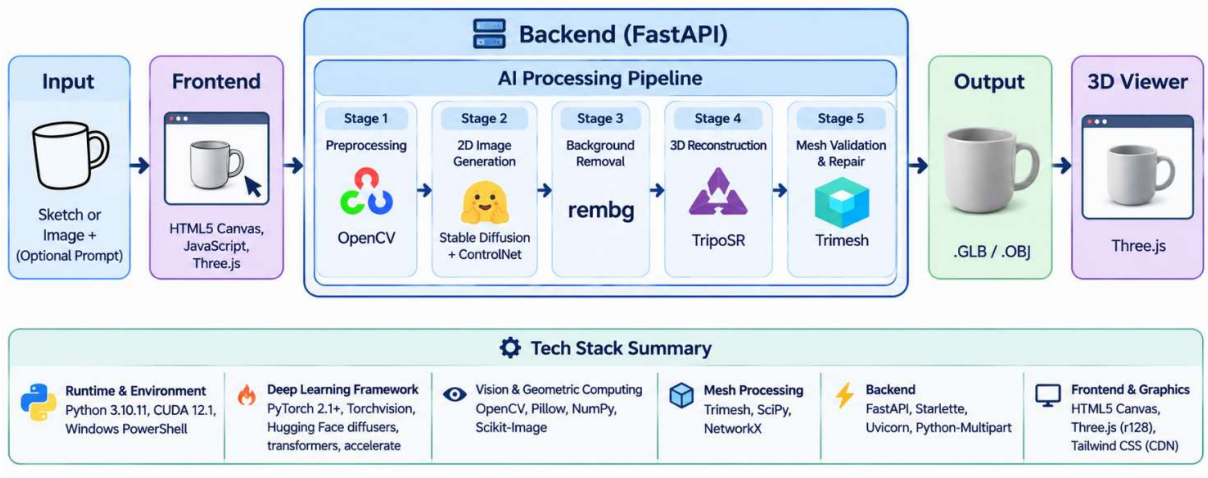
5. Texture and Appearance Enhancement: To generate suitable textures to improve the visual quality and realism of the 3D model.

6. 3D Model Preview: To provide an option to view and inspect the generated 3D model from different angles.

7. 3D File Export: To allow users to download or save the generated 3D model in commonly supported formats such as .glb or .obj.

8. Accessibility and Efficiency: To simplify the 3D modelling process and make 3D asset creation more accessible to students, designers, artists, and beginners without requiring advanced modelling skills.

#ARCHITECTURE:



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#DEPENDENCIES:

Software Requirements:

Software	Purpose
▪ Python 3.10 / compatible version	Main programming language
▪ Visual Studio Code	Code development and editing
▪ Git	Version control
▪ Windows / Linux	Operating system
▪ CUDA (if supported)	GPU acceleration
▪ NVIDIA GPU (if available)	AI model execution

Python Libraries and FrameworksL:

Library	Purpose
▪ PyTorch	Deep learning and AI model execution
▪ OpenCV	Image processing and sketch analysis
▪ NumPy	Numerical and array operations
▪ Pillow (PIL)	Image loading and manipulation
▪ Trimesh	3D mesh processing and validation
▪ rembg	Background removal
▪ Gradio	User interface development
▪ Transformers	Accessing pretrained AI models
▪ Diffusers	Stable Diffusion and ControlNet integration

AI Model Dependencies:

1. TripoSR Used for converting processed 2D images into 3D digital models.
2. Stable Diffusion Used for generating or refining visual representations based on the input sketch.
3. ControlNet

Hardware Requirements:

Component	Requirement
▪ Processor	Intel Core i5 / AMD Ryzen 5 or better
▪ RAM	16 GB recommended
▪ Storage	20 GB or more free space (depending on model weights)
▪ GPU	NVIDIA GPU recommended
▪ Internet	Required for downloading models and dependencies

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#INSTALLATION:

Requirements:- Windows 10/11 64-bit, Python 3.10.11, NVIDIA GPU with at least 6 GB VRAM and a CUDA 12.1-capable driver, about 7 GB free disk, internet on first run.

```
# 1. Get the code (use the actual repository URL)
git clone https://github.com/officegitshivam/Tech-zypher4.git
cd Tech-zypher4

# 2. Virtual environment
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip

# 3. PyTorch with CUDA (BEFORE the rest, at the versions pinned in
requirements.txt)
pip install torch==2.2.2 torchvision==0.17.2 torchaudio==2.2.2 --index-url
https://download.pytorch.org/whl/cu121

# 4. Remaining dependencies and TripoSR
pip install -r requirements.txt
pip install -e ./TripoSR        # see Section 15, item 3 if this fails

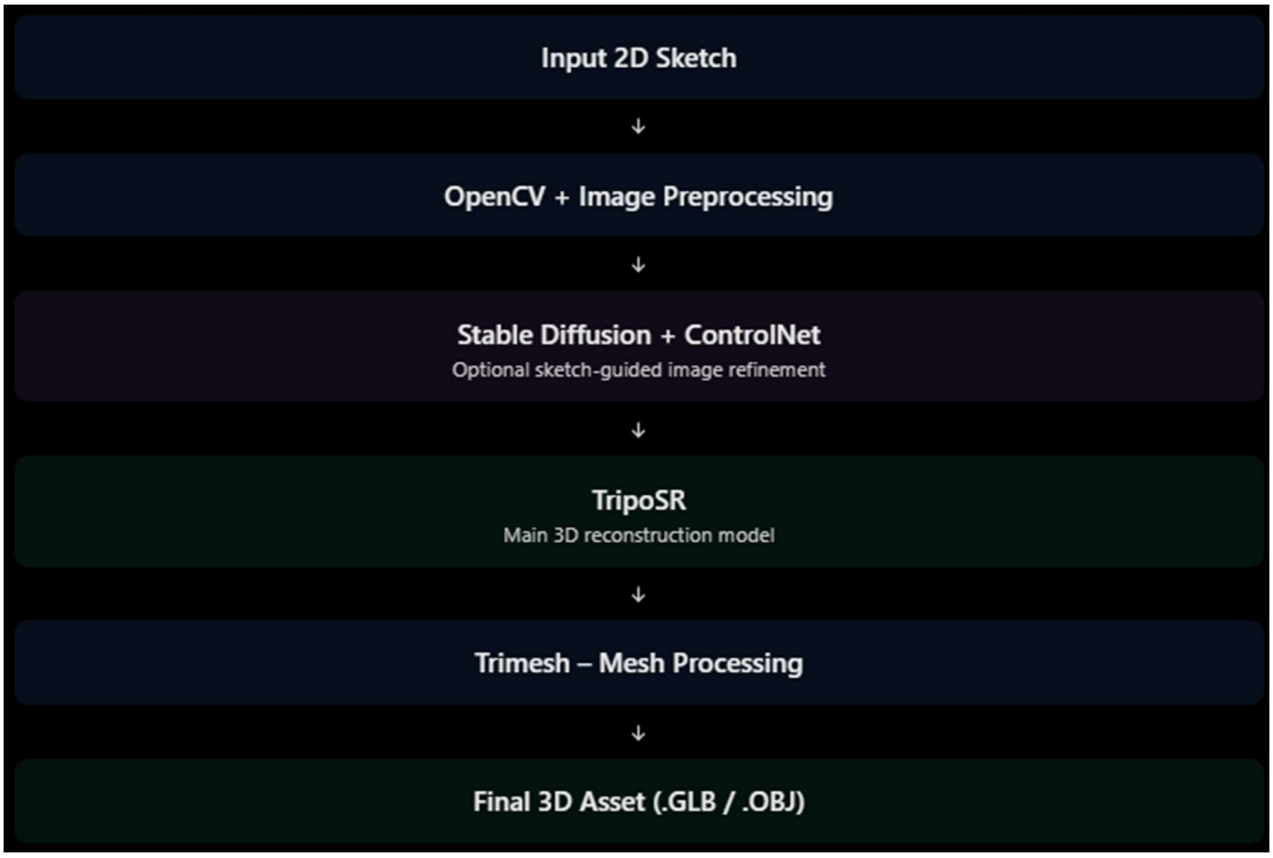
# 5. Start backend (terminal 1)
python server.py
# wait for: "Engines Loaded and Ready." and "Uvicorn running on
http://0.0.0.0:8000"

# 6. Start front end (terminal 2)
python -m http.server 5500
# open http://127.0.0.1:5500/interface.html
```

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#MODEL SELECTION:



#EXPECTED INPUT-OUTPUT:

INPUT: drawn sketch (canvas), or .png / .jpg / .jpeg image; optional free-text prompt; optional control weight.

OUTPUT: latest_contours.png, latest_render.png, latest_isolated.png, reconstructed_asset.glb, reconstructed_asset.obj, plus the validation report in the API response.

TIME-REQUIRED: Average time of generation of 3D image is 3 Minutes.

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#HARDWARE AND RESOURCES:

TEST-MACHINE: Windows 11, NVIDIA GeForce RTX 4050 Laptop GPU (6 GB), 16 GB RAM, Intel Core i5 / AMD Ryzen 7 class CPU.

Step	Module	Avg time	Peak VRAM
1. Contours	vision	~0.8 s	0 (CPU)
2. Diffusion	diffusion	~2.3mins	~4.4 GB
3. Matting	segmentation	~4.5 s	~0.6 GB
4. Reconstruction	reconstruction	~20 s	~4.8 GB
5. Repair/export	validator	~1.5 s	0 (CPU)
Total	end to end	~3minutes	~5.1 GB peak

ISSUES AND LIMITATIONS:

1.Single-View Ambiguity: While TripoSR infers plausible occluded surfaces, the back side of complex asymmetric objects (e.g., a chair with intricate rear carvings) is estimated based on learned priors rather than ground-truth visibility.

2.Thin Geometry Quantization: At a marching cubes resolution of 128^3 ultra-thin features (such as thin antennae, needles, or paper edges) can experience minor topological bridging.

3.GPU VRAM Ceiling: Running concurrent background processes that consume VRAM (e.g., local LLMs or games) may push memory usage beyond 6 GB, triggering a PyTorch OOM exception during the diffusion-to-reconstruction handoff.

4.LIVE SERVER ISSUE: Live server often serves the project from a sub-folder path or appends query params. Constructing a url or navigating directly while the modal/form event is firing triggers Live Server's reload listener before browser finishes navigation.